4-Year Computational Astrophysics

Master Checklist — Revised Edition

Python + C++ + Physics + Astrophysics + Real Data · Lund University 2027

Plan overview

Metric	Value	Time split	Core identity
Weekly target	14–18 hours (min 14, strong 16–18)	Physics/Astro 50% · Programming 30% · Math/Docs 20%	Scientific software builder — Python for analysis, C++ for performance

Annual project ladder

Year	Theme	End-of-year project	Key new capability
Year 1	Foundations	2D Orbital Mechanics and Gravity Simulation Suite	Scientific Python · Numerical methods · C++ basics · Classical mechanics
Year 2	Intermediate + Real Data	Exoplanet/Orbital Data Analysis + Simulation Toolkit	Real Kepler data · Fourier analysis · EM basics · N-body · Research writing
Year 3	Advanced Simulation	High-Performance Astrophysical Simulation Engine	Advanced C++ · Intro quantum mechanics · Performance profiling · Real survey data
Year 4	Flagship Platform	Flagship Computational Astrophysics Platform	Product architecture · Portfolio packaging · Monetization · Presentation

What changed from the original plan — and why

All original content is preserved. Items marked ★ NEW add missing mathematics and physics. Items marked ↻ UPDATED pivot synthetic-data projects to real astrophysical datasets.

Symbol	Meaning	Where it appears	Why it was added
★ NEW	New task added	Y1 M1,M2,M3 · Y2 M1,M2,M3,M9 · Y3 M4	Fills mathematical gaps (Fourier, probability, complex numbers, EM, quantum) identified as missing for Lund's BSc curriculum
↻ UPDATED	Task or project changed	Y2 M6, Y3 M8, Y4 M1	Pivots synthetic-data projects to use real astrophysical datasets (Kepler, Gaia, SDSS) — the single most important portfolio upgrade

Real datasets used in this plan

All datasets below are free and publicly accessible. No institutional access required.

Dataset	Source	What it contains	Used in	Free?
NASA Exoplanet	exoplanetarchive.ipac.calt	Kepler/TESS transit light	Year 2 Month 6	**Yes**

Dataset	Source	What it contains	Used in	Free?
Archive	ech.edu	curves, orbital parameters for all confirmed exoplanets		
Gaia DR3	gea.esac.esa.int	Position, proper motion, parallax, and photometry for 1.8 billion stars	Year 2 Month 9	**Yes**
SDSS (Sloan Digital Sky Survey)	skyserver.sdss.org	Galaxy photometry, spectra, and morphology for millions of objects	Year 3 Month 8	**Yes**
Open Exoplanet Catalogue	github.com/ openexoplanetcatalogue	Orbital parameters, mass, radius for all confirmed exoplanets in CSV format	Year 2 Month 8, Year 4	**Yes**

Year 1 — Foundations of Scientific Computing and Classical Physics

Weekly target: 14–18 hours · End-of-year project: 2D Orbital Mechanics and Gravity Simulation Suite
New this year: complex numbers, probability basics, and intro electrostatics — foundational for all later astrophysics work.

Year 1 — Month 1 · Scientific Python Reset + Math Refresh

Monthly project: Scientific Python Toolkit

- [ ] Deeply learn NumPy arrays, slicing, broadcasting, and vectorized thinking
- [ ] Learn Matplotlib plotting, labels, legends, and multi-plot workflows
- [ ] Refresh vectors, trigonometry, functions, and graph interpretation
- [ ] Refresh derivative intuition and basic calculus manipulation
- [ ] Introduce complex numbers: i, Euler's formula $e^{i\theta} = \cos\theta + i\cdot\sin\theta$, and polar form ★ **NEW**
- [ ] Create a clean project folder system and Git/GitHub workflow
- [ ] Start a physics-programming journal for notes, formulas, and mistakes

Checkpoint — Date completed: _____	Notes / blockers: _____

Year 1 — Month 2 · Numerical Thinking in Python

Monthly project: 1D Kinematics Solver

- [ ] Study floating-point error and why numerical results can drift
- [ ] Compare loops vs vectorization in speed and readability
- [ ] Implement finite-difference derivatives
- [ ] Implement simple numerical integration routines
- [ ] Introduce probability: random variables, mean, variance, normal distribution — implement in NumPy ★ **NEW**
- [ ] Apply descriptive statistics to a small dataset (e.g. star catalogue CSV) — mean, std, histogram ★ **NEW**
- [ ] Review displacement, velocity, and acceleration relationships
- [ ] Build and test a 1D motion solver with plots

Checkpoint — Date completed: _____	Notes / blockers: _____

Year 1 — Month 3 · Mechanics I

Monthly project: Force Simulator

- [ ] Study Newton's laws deeply and solve force problems by hand
- [ ] Practice free-body diagrams until they feel automatic
- [ ] Work on friction, tension, and inclined plane problems
- [ ] Convert force-balance equations into code
- [ ] Use dimensional analysis and units carefully in every problem
- [ ] Intro electrostatics: Coulomb's law, electric field concept, and force-at-a-distance intuition — compare to gravity ★ **NEW**

Checkpoint — Date completed: _____	Notes / blockers: _____

Year 1 — Month 4 · C++ from Zero

Monthly project: C++ Physics Calculator Pack

- [ ] Set up a C++ compiler, editor, and build workflow
- [ ] Learn syntax, variables, conditionals, loops, and functions
- [ ] Learn vectors, arrays, and standard input/output
- [ ] Write small console-based physics calculators
- [ ] Recreate simple Python programs in C++

Checkpoint — Date completed: _____	Notes / blockers: _____

Year 1 — Month 5 · Energy, Work, Momentum

Monthly project: Collision and Energy Analyzer

- [ ] Study kinetic energy, potential energy, work, and conservation laws
- [ ] Study momentum, impulse, and elastic/inelastic collisions
- [ ] Solve representative textbook-style problems manually
- [ ] Code energy and momentum trackers
- [ ] Visualize conservation and loss scenarios

Checkpoint — Date completed: _____	Notes / blockers: _____

Year 1 — Month 6 · Differential Equations and ODE Thinking

Monthly project: ODE Playground

- [ ] Understand why many physical systems are differential equations
- [ ] Implement Euler and Euler-Cromer methods
- [ ] Simulate free-fall with drag and spring-mass systems
- [ ] Compare analytic and numerical results
- [ ] Track numerical error and stability

Checkpoint — Date completed: _____	Notes / blockers: _____

Year 1 — Month 7 · Gravitation Basics

Monthly project: Orbit Basics Calculator

- [ ] Study Newtonian gravitation and inverse-square behavior
- [ ] Learn orbital speed, period, circular motion, and escape velocity
- [ ] Compute satellite examples by hand and by code
- [ ] Visualize gravitational field and potential curves

Checkpoint — Date completed: _____	Notes / blockers: _____

Year 1 — Month 8 · Data Handling and Intro Analysis

Monthly project: Mini Astrophysics Data Notebook

- [ ] Learn Pandas basics and CSV data workflows
- [ ] Clean, inspect, and summarize small datasets
- [ ] Learn simple regression and trend fitting
- [ ] Plot noisy data and interpret residuals

[ ] Write short notebook-style analysis summaries

Checkpoint — Date completed: _____	Notes / blockers: _____

Year 1 — Month 9 · Better C++ + OOP Basics

Monthly project: C++ Body Simulation Core

[ ] Learn structs, classes, constructors, and header/source separation
[ ] Create reusable Vector2 and Body classes
[ ] Practice debugging and file organization
[ ] Port an earlier physics project from Python to C++

Checkpoint — Date completed: _____	Notes / blockers: _____

Year 1 — Month 10 · Numerical Integration for Orbits

Monthly project: Integrator Comparison Lab

[ ] Learn why naive Euler is weak for orbital systems
[ ] Implement Euler-Cromer, Verlet, and RK4
[ ] Compare integrators on the same orbit setup
[ ] Measure stability and energy drift
[ ] Document which integrator works best and why

Checkpoint — Date completed: _____	Notes / blockers: _____

Year 1 — Month 11 · Visualization and Interactive Simulation

Monthly project: Interactive Orbit Viewer v1

[ ] Learn animation with Matplotlib or Pygame
[ ] Turn static orbit calculations into animated systems
[ ] Add adjustable initial conditions and trails
[ ] Improve code modularity and simulation structure

Checkpoint — Date completed: _____	Notes / blockers: _____

Year 1 — Month 12 · Year 1 Integration

Monthly project: Year 1 Final Project Build

[ ] Refactor reusable code modules
[ ] Build a clean 2-body and 3-body gravity simulator
[ ] Add conservation checks, plots, and parameter controls
[ ] Write a short technical report with equations and results
[ ] Publish the project neatly on GitHub

Checkpoint — Date completed: _____	Notes / blockers: _____

[ ] Year 1 Complete — 2D Orbital Mechanics and Gravity Simulation Suite published on GitHub

Year 2 — Intermediate Computational Physics and Real Astrophysics Data

Weekly target: 15–18 hours · End-of-year project: Exoplanet/Orbital Data Analysis + Simulation Toolkit
New this year: eigenvalues, Fourier analysis, EM basics, Doppler shift, real Kepler light curves, Gaia stellar data.

Year 2 — Month 1 · Calculus and Linear Algebra for Physics

Monthly project: Linear Algebra for Physics Notebook

- [ ] Review multivariable functions and partial derivatives
- [ ] Learn matrices, vector fields, and coordinate intuition
- [ ] Use NumPy linear algebra tools confidently
- [ ] Connect matrix thinking to physical systems
- [ ] Study eigenvalues and eigenvectors as physical concepts — normal modes, principal axes, diagonalisation ★ **NEW**
- [ ] Implement a simple eigenvalue solver in Python and apply it to a 2D oscillator or rotation problem ★ **NEW**

Checkpoint — Date completed: _____	Notes / blockers: _____

Year 2 — Month 2 · C++ Intermediate + Fourier Analysis

Monthly project: C++ Data Logger + Fourier Toolkit

Original: C++ Data Logger + Simulation Runner. Fourier analysis added alongside C++ work.

- [ ] Learn pointers and references correctly
- [ ] Understand stack vs heap at a practical level
- [ ] Learn file input/output and simulation logging
- [ ] Benchmark simple C++ programs
- [ ] Improve code organization and debugging habits
- [ ] Study Fourier series: decomposing periodic signals into sine/cosine components ★ **NEW**
- [ ] Implement a Discrete Fourier Transform (DFT) in Python from scratch — no library first ★ **NEW**
- [ ] Apply FFT (numpy.fft) to a light-curve-like signal — identify periodicity and noise ★ **NEW**
- [ ] Understand why Fourier analysis appears in spectroscopy, power spectra, and period-finding ★ **NEW**

Checkpoint — Date completed: _____	Notes / blockers: _____

Year 2 — Month 3 · Oscillations, Waves, and EM Basics

Monthly project: Oscillation and Wave Simulation Lab

Original: Oscillations and Waves (mechanical only). Electromagnetic waves added — essential for astrophysics.

- [ ] Study SHM, damping, driven oscillations, and resonance
- [ ] Plot phase-space trajectories
- [ ] Simulate resonance and compare parameter changes
- [ ] Link oscillation behavior to broader physical systems
- [ ] Study electromagnetic waves: E and B fields, speed of light, wave equation conceptually ★ **NEW**
- [ ] Understand electromagnetic spectrum and why different wavelengths reveal different astrophysical

phenomena ★ NEW

[ ] Study the Doppler shift for light — derive the formula and apply it to stellar radial velocity ★ NEW

[ ] Code a Doppler-shift calculator for spectral lines — this directly supports your exoplanet work ★ NEW

Checkpoint — Date completed: _____	Notes / blockers: _____

Year 2 — Month 4 · Thermodynamics and Statistical Thinking

Monthly project: Ideal Gas and Distribution Simulator

[ ] Study temperature, heat, ideal gas behavior, and entropy basics
[ ] Learn microscopic interpretations and distribution intuition
[ ] Code simple particle-speed or gas-behavior models
[ ] Relate thermodynamic ideas to stellar matter intuition

Checkpoint — Date completed: _____	Notes / blockers: _____

Year 2 — Month 5 · Intro Astrophysics I

Monthly project: Stellar Property Calculator

[ ] Study luminosity, flux, magnitude, radius, and temperature relations
[ ] Learn the inverse-square light law and Stefan-Boltzmann law
[ ] Study the meaning of the H-R diagram
[ ] Code stellar property calculators and plots

Checkpoint — Date completed: _____	Notes / blockers: _____

Year 2 — Month 6 · Astronomy Data Analysis — Real Kepler Data

Monthly project: Light Curve Analyzer v1 (Real Data)

UPDATED: This project now uses real Kepler transit light curves from the NASA Exoplanet Archive instead of synthetic data. This is the most important portfolio upgrade in the plan.

[ ] Download a real Kepler light curve from the NASA Exoplanet Archive (exoplanetarchive.ipac.caltech.edu) ↻ UPDATED

[ ] Load and inspect the FITS or CSV file — understand the column structure and time/flux format ↻ UPDATED

[ ] Clean the data: identify gaps, outliers, and instrumental systematics ↻ UPDATED

[ ] Detect the transit dip using a box least squares (BLS) algorithm or simple threshold detection ↻ UPDATED

[ ] Measure transit depth, duration, and period from real observations ↻ UPDATED

[ ] Estimate the planet-to-star radius ratio from the transit depth ↻ UPDATED

[ ] Apply residuals analysis and uncertainty estimation to real noisy measurements
[ ] Write a notebook-style analysis report with your findings on the real system

[ ] Compare your measured period to the known value in the Exoplanet Archive — quantify your error ↻ UPDATED

Checkpoint — Date completed: _____	Notes / blockers: _____

Year 2 — Month 7 · Numerical Methods II

Monthly project: Numerical Methods Toolkit

- [ ] Implement root finding with bisection and Newton-Raphson
- [ ] Learn interpolation and numerical quadrature
- [ ] Use optimization ideas for parameter fitting
- [ ] Apply methods to astrophysical equations

Checkpoint — Date completed: _____	Notes / blockers: _____

Year 2 — Month 8 · Gravitation II and N-body

Monthly project: Three-Body Exploration Simulator

- [ ] Study the three-body problem and barycenter intuition
- [ ] Simulate restricted 3-body cases
- [ ] Explore sensitivity to initial conditions
- [ ] Study Lagrange point ideas conceptually
- [ ] Download orbital parameters for real exoplanet systems from Open Exoplanet Catalogue — compare to your simulated orbits ★ **NEW**

Checkpoint — Date completed: _____	Notes / blockers: _____

Year 2 — Month 9 · Galaxies, Cosmology, and Real Stellar Data

Monthly project: Galaxy Rotation and Stellar Data Notebook

Extended: Gaia DR3 real stellar data added alongside galaxy rotation work.

- [ ] Study galaxy structure and the rotation-curve problem
- [ ] Learn redshift and Hubble's law basics
- [ ] Plot synthetic rotation curves
- [ ] Compute distance-redshift examples and interpretations
- [ ] Download a subset of Gaia DR3 data (gea.esac.esa.int) — parallax and proper motion for a star cluster ★ **NEW**
- [ ] Convert Gaia parallax to distance and plot the cluster's sky distribution ★ **NEW**
- [ ] Compare your cluster distance estimate to published values ★ **NEW**

Checkpoint — Date completed: _____	Notes / blockers: _____

Year 2 — Month 10 · Research and Technical Writing

Monthly project: Mini Research Report Portfolio

- [ ] Read beginner-friendly computational physics or astrophysics papers
- [ ] Summarize methods, assumptions, and conclusions
- [ ] Learn intro-method-results-discussion structure
- [ ] Improve reproducibility and documentation habits

Checkpoint — Date completed: _____	Notes / blockers: _____

Year 2 — Month 11 · Tool Consolidation

Monthly project: Annual Project Prototype

- [ ] Merge the strongest code from Years 1 and 2
- [ ] Choose the final annual project architecture
- [ ] Build a clean data pipeline and simulation pipeline
- [ ] Upgrade visuals and outputs

Checkpoint — Date completed: _____	Notes / blockers: _____

Year 2 — Month 12 · Year 2 Integration

Monthly project: Year 2 Final Project Build

- [ ] Finalize a toolkit that reads real orbital or light-curve data
- [ ] Estimate parameters from data using your numerical methods
- [ ] Simulate the corresponding system
- [ ] Visualize results cleanly with publication-style figures
- [ ] Write a polished technical report in intro-method-results-discussion format

Checkpoint — Date completed: _____	Notes / blockers: _____

[ ] Year 2 Complete — Exoplanet/Orbital Data Analysis Toolkit (using real Kepler data) published on GitHub

Year 3 — Advanced Astrophysics and High-Performance Simulation

Weekly target: 15–19 hours · End-of-year project: High-Performance Astrophysical Simulation Engine
New this year: intro quantum mechanics, real SDSS/Gaia observational comparison.

Year 3 — Month 1 · Advanced C++ for Scientific Computing

Monthly project: C++ Performance Rewrite

- [ ] Improve const correctness and class design
- [ ] Refactor earlier C++ code to a cleaner standard
- [ ] Learn profiling basics and identify bottlenecks
- [ ] Compare Python and C++ runtime behavior carefully

Checkpoint — Date completed: _____	Notes / blockers: _____

Year 3 — Month 2 · Linear Algebra and State Modeling

Monthly project: State-Space Physics Toolkit

- [ ] Work more deeply with state vectors and matrix representations
- [ ] Learn coordinate transformations for modeled systems
- [ ] Build utilities for representing system states numerically

Checkpoint — Date completed: _____	Notes / blockers: _____

Year 3 — Month 3 · Advanced Numerical Methods

Monthly project: Adaptive Integrator Benchmark Suite

- [ ] Study adaptive step-size ideas and error estimation
- [ ] Benchmark harder orbital cases
- [ ] Compare long-run stability across methods
- [ ] Improve integrator selection logic

Checkpoint — Date completed: _____	Notes / blockers: _____

Year 3 — Month 4 · Stellar Structure, Dynamics, and Intro Quantum Mechanics

Monthly project: Simple Stellar Structure + Quantum Intro

Quantum mechanics introduction added. You will encounter this in Lund Year 1 — arriving with zero exposure is a significant disadvantage.

- [ ] Study hydrostatic equilibrium intuition
- [ ] Learn mass-radius-luminosity relationships
- [ ] Build simple stellar structure approximations in code
- [ ] Document assumptions and limits clearly
- [ ] Study wave-particle duality and the de Broglie wavelength conceptually ★ **NEW**
- [ ] Implement the particle-in-a-box solution in Python — plot wavefunctions and energy levels ★ **NEW**

[ ] Simulate the quantum harmonic oscillator — compare to classical SHM ★ NEW

[ ] Understand why quantum mechanics governs stellar spectra (atomic energy levels, photon emission) ★ NEW

[ ] Read: how spectral lines are produced — link quantum transitions to observational astronomy ★ NEW

Checkpoint — Date completed: _____	Notes / blockers: _____

Year 3 — Month 5 · Galactic Motion

Monthly project: Galaxy Rotation Curve Fitter

[ ] Study galaxy mass distribution ideas
[ ] Model visible vs inferred mass behavior
[ ] Simulate and fit galaxy rotation curves
[ ] Compare curve families and parameter effects

Checkpoint — Date completed: _____	Notes / blockers: _____

Year 3 — Month 6 · Scientific Software Design

Monthly project: Scientific Engine Skeleton

[ ] Learn modular architecture, testing, and reproducibility
[ ] Add config systems, logging, and reusable modules
[ ] Create a maintainable engine skeleton

Checkpoint — Date completed: _____	Notes / blockers: _____

Year 3 — Month 7 · Performance Expansion

Monthly project: Optimized Large-System Simulator v1

[ ] Profile bottlenecks in large simulations
[ ] Reduce allocations and improve data structures
[ ] Explore basic multicore or parallel ideas if manageable
[ ] Benchmark larger systems systematically

Checkpoint — Date completed: _____	Notes / blockers: _____

Year 3 — Month 8 · Observational Comparison with Real Survey Data

Monthly project: Simulation vs Real Data Report

UPDATED: Now uses real SDSS or Gaia survey data instead of synthetic observational-style data. This is what separates a portfolio project from a classroom exercise.

[ ] Choose a real dataset: SDSS galaxy photometry (skyserver.sdss.org) or Gaia stellar kinematics ↻ UPDATED

[ ] Download and load the dataset — inspect its structure, units, and coverage ↻ UPDATED

[ ] Compare your simulation output to a relevant observable quantity in the real data ↻ UPDATED

[ ] Measure fit quality and residuals between model and real observations ↻ UPDATED

[ ] Document assumptions, limitations, and what physical effects your model ignores
[ ] Write a research-style comparison report with figures and quantified discrepancies

Checkpoint — Date completed: _____	Notes / blockers: _____

Year 3 — Month 9 · Visualization and Presentation Layer

Monthly project: Astrophysics Visualization Suite

- [ ] Build polished plotting utilities and playback visuals
- [ ] Add scenario controls or parameter presets
- [ ] Prepare figures that are presentation-ready

Checkpoint — Date completed: _____	Notes / blockers: _____

Year 3 — Month 10 · Research-Style Investigation

Monthly project: Mini Computational Astrophysics Study

- [ ] Choose one focused simulation question
- [ ] Run controlled experiments and record methodology
- [ ] Interpret findings with care
- [ ] Write a research-style document

Checkpoint — Date completed: _____	Notes / blockers: _____

Year 3 — Month 11 · Annual Project Assembly

Monthly project: Annual Engine Candidate Build

- [ ] Merge optimized C++ core with Python front end
- [ ] Add diagnostics, exports, and scenario presets
- [ ] Prepare strong documentation and demo material

Checkpoint — Date completed: _____	Notes / blockers: _____

Year 3 — Month 12 · Year 3 Integration

Monthly project: Year 3 Final Project Build

- [ ] Finalize a high-performance simulation engine
- [ ] Include benchmark results and diagnostics
- [ ] Polish documentation and repo structure
- [ ] Prepare a demo-ready version

Checkpoint — Date completed: _____	Notes / blockers: _____

[ ] Year 3 Complete — High-Performance Simulation Engine with real data comparison published on GitHub

Year 4 — Flagship Platform, Portfolio, and Monetization

Weekly target: 14–20 hours · End-of-year project: Flagship Computational Astrophysics Platform
Flagship must include at least one real-data pipeline. This is what separates a portfolio platform from a simulation demo.

Year 4 — Month 1 · Choose Flagship Direction

Monthly project: Flagship Project Blueprint

- [ ] Review all previous projects and choose one flagship domain
- [ ] Define target users: universities, students, educators, or judges
- [ ] Write a project vision, scope, and feature hierarchy
- [ ] Define MVP vs advanced features clearly
- [ ] Audit which projects used real data — plan to include at least one real-data pipeline in the flagship ↻ **UPDATED**
- [ ] List datasets the flagship will ingest (Kepler, Gaia, SDSS, or Open Exoplanet Catalogue) ↻ **UPDATED**

Checkpoint — Date completed: _____	Notes / blockers: _____

Year 4 — Month 2 · Product Architecture

Monthly project: Architecture and Technical Design Document

- [ ] Design modules for engine, data, UI, visualization, and docs
- [ ] Decide what remains in Python and what runs in C++
- [ ] Design a professional repository structure

Checkpoint — Date completed: _____	Notes / blockers: _____

Year 4 — Month 3 · Core Engine Upgrade

Monthly project: Flagship Engine v1

- [ ] Refactor the simulation engine for extensibility and stability
- [ ] Improve performance and large-scenario handling
- [ ] Prepare strong testing scenarios

Checkpoint — Date completed: _____	Notes / blockers: _____

Year 4 — Month 4 · Analysis Layer

Monthly project: Scientific Analysis Module

- [ ] Build robust analysis notebooks or modules
- [ ] Add diagnostics, fitting, and error analysis
- [ ] Export polished graphs and tables

Checkpoint — Date completed: _____	Notes / blockers: _____

Year 4 — Month 5 · UI and Interaction Layer

Monthly project: Interactive Astrophysics Interface

[ ] Add interactive controls and scenario presets
[ ] Make the system understandable to other users
[ ] Improve usability and explanatory flow

Checkpoint — Date completed: _____	Notes / blockers: _____

Year 4 — Month 6 · Physics Depth Expansion

Monthly project: Advanced Physics Extension Module

[ ] Add one advanced physical extension
[ ] Examples: perturbations, mass profiles, observational comparison, or advanced integration
[ ] Validate the extension and document limitations

Checkpoint — Date completed: _____	Notes / blockers: _____

Year 4 — Month 7 · Portfolio Packaging

Monthly project: Public Portfolio Package

[ ] Write technical and beginner-friendly docs
[ ] Create diagrams, screenshots, and demo material
[ ] Prepare a portfolio-quality README and project narrative

Checkpoint — Date completed: _____	Notes / blockers: _____

Year 4 — Month 8 · Research or Competition Version

Monthly project: Research Submission Draft

[ ] Prepare a research-style summary
[ ] State question, method, result, and limitations
[ ] Produce polished figures and a presentation outline

Checkpoint — Date completed: _____	Notes / blockers: _____

Year 4 — Month 9 · Monetization Exploration

Monthly project: Monetizable MVP Concept

[ ] Choose realistic monetization routes
[ ] Create one small offerable or sellable version
[ ] Define audience, value proposition, and pricing hypothesis

Checkpoint — Date completed: _____	Notes / blockers: _____

Year 4 — Month 10 · Testing and Refinement

Monthly project: Release Candidate

[ ] Stress-test the software and fix issues
[ ] Improve onboarding, usability, and documentation
[ ] Record known limitations honestly

Checkpoint — Date completed: _____	Notes / blockers: _____

Year 4 — Month 11 · Final Assembly

Monthly project: Flagship vFinal Beta

[ ] Merge all polished modules into one coherent product
[ ] Finalize report, repo structure, and explanation formats
[ ] Prepare a 5-minute and 15-minute presentation version

Checkpoint — Date completed: _____	Notes / blockers: _____

Year 4 — Month 12 · Year 4 Integration

Monthly project: Year 4 Final Project Build

[ ] Finalize the flagship computational astrophysics platform
[ ] Ensure it includes simulation, analysis, visualization, and documentation
[ ] Confirm at least one end-to-end real dataset pipeline is working and documented ↻ **UPDATED**
[ ] Prepare demo examples and a monetization-path explanation
[ ] Package it as a university-ready and portfolio-ready showcase

Checkpoint — Date completed: _____	Notes / blockers: _____

[ ] Year 4 Complete — Flagship Computational Astrophysics Platform published and portfolio-ready

Monthly completion rules

- [ ] Do not move to the next month until the current month's mini-project is finished.
- [ ] Each month should produce at least one working artifact: notebook, simulator, report, or code module.
- [ ] Every project should be saved with versioned folders and a short summary of what you learned.
- [ ] At the end of every 3 months, review mistakes, unfinished tasks, and weak topics before continuing.
- [ ] During exam-heavy periods, keep progress alive with a 12–14 hours/week maintenance mode.

Personal weekly log template

Week	Hours done	Main topic	What I finished
Week 1			
Week 2			
Week 3			
Week 4			
Week 5			
Week 6			